

Demonstration of Proposed Features

Date	10/07/2026
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Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document presents the major features proposed during the planning phase of the **Voice-Based Concept Understanding Analyser (VBCUA)** project and indicates whether each feature has been successfully implemented and demonstrated.

The application is developed using **Python** and **Streamlit** and integrates **OpenAI Whisper**, **Sentence-BERT**, and **Librosa** to evaluate conceptual understanding and spoken communication. It provides semantic similarity analysis, speech fluency assessment, AI-based scoring, waveform visualization, and downloadable PDF reports.

S.No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrated (Yes / No)	Remarks
1	Audio Upload Module	Uploads user audio (.wav/.mp3) for analysis.	Implemented	Yes	Supports multiple audio formats.
2	Speech-to-Text Transcription	Converts speech into text using OpenAI Whisper..	Implemented	Yes	Accurate transcription for English speech..
3	Semantic Concept Evaluation	Compares transcript with reference concepts using Sentence-BERT embeddings.	Implemented	Yes	Calculates semantic similarity score..
4	Speech Fluency Analysis	Detects filler words, pause ratio, RMS	Implemented	Yes	Instant result shown to the user.

		energy, and speaking clarity using Librosa..			
5	Interactive Dashboard	Displays transcript, waveform, scores, graphs, and qualitative feedback.	Implemented	Yes	Built with Streamlit for interactive visualization...
6	PDF Report Generation	Generates downloadable PDF reports containing evaluation metrics and AI feedback..	Implemented	Yes	Reports are ready for academic review and progress tracking.

Feature Implementation Summary:

Metric	Count / Value
Total Features Proposed	6
Total Features Implemented	6
Total Features Demonstrated	6
Overall Implementation Rate (%)	100%